

Call for Expressions of Interest	
<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Universal Sensor-to-Game Interface
<i>Company:</i>	Please provide company name
Contact Details	
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<i>E-mail:</i>	Additional contact e-mail
Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☒ If Applicable: Some of the hardware is currently being built and I don't expect this group to work on its development. However, if it is the interest of the group or individual members, involvement in it may be arranged.
<i>Relevant research areas (if applicable)¹:</i>	Assistive technologies, cyber-physical health, mixed reality, game development, HCI.
Project Background Information	
Modern sensing technologies such as IMUs, EMG, gesture recognition and motion capture can provide new ways of interacting with computers and games. This is a current topic in immersive gaming, and particularly important in accessibility and assistive technologies. However, applications often need to be specifically adapted to each sensing technology. This project aims to develop a flexible software platform that acts as a bridge between sensor-based human input and interactive applications. The system will receive inputs from different sources and translate them into standardised controls such as buttons, analogue sticks and triggers. Users should be able to configure these mappings without modifying the application itself.	

¹ An overview of different research areas in the school of computer science can be found [here](#).

The focus of the project is the design and development of the software infrastructure connecting sensing systems to interactive applications, rather than the development of new sensor-processing algorithms or a complete game.

Requirements

The group must provide:

Students will work collaboratively to design and develop a software solution with the following key requirements:

Input Integration: Support simulated, recorded and at least one real or external input source, such as an IMU or gesture-recognition system.

Input Mapping: Provide a configurable mechanism for mapping sensor or recognised movements to standard controls such as buttons, analogue sticks and triggers.

Real-Time Processing: Process incoming input streams with sufficiently low latency for interactive use and provide visual feedback on the generated controls.

User Interface: Develop an intuitive interface for configuring input sources, mappings and system settings, including saving and loading configurations.

Application Integration: Provide controller or equivalent output that can be used by an existing application or game without requiring changes to that application.

The system should use a modular architecture so that new input sources and output methods can be added in the future.

Skills

Highly desirable:

- Proficiency in Python or another programming language
- Knowledge of software design and object-oriented programming
- Experience developing graphical user interfaces
- Knowledge of communication protocols

Desirable:

- Experience with real-time or event-driven systems
- Familiarity with APIs or external software interfaces
- Basic understanding of sensor technologies
- Experience with version control and software testing

Submission / Award information

Date of Submission:

September 2026

Date of Pitch:

October 2026

<i>Notification of award:</i>	October 2026
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For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.